University of Cape Town
Department of Computer Science

Computer Science CSC3003S
Concession Test, 2018

Answer all questions.
Approximate marks per question are shown in brackets.

Time: 40 minutes **Marks: 40**

Section 1. Advanced Software Design 1 [21 Marks]

Question 1 [6 Marks]

a. What is test driven development? Highlight how it differs from more traditional software development methods. [5]

Test Driven Development (TDD) can be described as test first development + refactoring [1]. So TDD turns traditional development around and instead of writing functional code first and then your testing code as an afterthought [1], you first write your test code before your functional code [1]. In addition, the development occurs in very small steps: one test and a small bit of code at a time [1]. Once the code passes the test (i.e. it works), it is refactored to ensure that it's remains of high quality [1]. After refactoring the tests are of course applied again before moving on [1].

The radical point is that new code is only written when an automated test fails [1].

b. Which of the following is not a principle of agile software development methods: [1]

i. The team must be empowered to make decisions
ii. Requirements are fixed
iii. Develop small, incremental releases and iterate
iv. Focus on frequent delivery of products
v. Apply the 80/20 rule

ii – Requirements are fixed

Question 2 [4 Marks]

a. In Scrum, there are only three roles. One of these is the Scrum Master who is responsible for the Scrum process and ensuring everyone follows Scrum rules and practices. Name the other two roles are briefly discuss what they do. [4]

Product owner *[1]. Represent the interests of everyone with a stake in the project and its resulting system [1]. Product Owner uses the Product Backlog to ensure the most valuable functionality is produced first and built upon. Achieves initial and ongoing funding.*

Creates

- the project's initial overall requirements

- return on investment (ROI) objectives

- release plans.

Scrum Team [1]. *Responsible for developing functionality [1]. Responsible for deciding how to turn Product Backlog into an increment of functionality within an iteration. Are self-managing, self-organizing, and cross-functional.*

Question 3 [11 Marks]

a. Briefly explain what a software design pattern is [1]

"Each pattern is a three-part rule, which expresses a relation between a certain context, a problem and a solution."

OR Definition*: A pattern is a solution to a problem in a context.*

[1 mark]

b. Indicate two benefits of design patterns? [2]

Capture expertise and make it accessible to non-experts in an encapsulated design pattern

Help communication amongst developers by providing a common language

Improve design understandability

Make it easier to reuse successful designs and avoid alternatives that diminish reusability

Facilitate design modifications

The design is more easily understood

Improve design documentation

//any two [1 mark each]

c. What is the advantage of defining a software component as a particular type? [1]

allows it to be easily substituted/replaced with a component of the same type.

d. Indicate two advantages and one disadvantages of a layered architecture. [3]

Advantages:

System easier to comprehend: each layer has specific purpose

Facilitates 'high-cohesion' within layers and 'low-coupling' between layers

Prevents source code changes from rippling throughout system

Code changes should be confined within a specific layer

Minimizing coupling makes it easier to swap out a technical service in a lower layer and replace it with another technology

Replacing RDBMS without major code re-write in application logic

Lower layers contain more reusable functionality

//any two [1 mark each]

Disadvantages:

hampered optimisation, functions of each layer are carried out completely before protocol data unit is passed to the next layer,

Increased layering involves increased complexity (via inter-layer dependencies)

//any one [1 mark]

e. An online e-commerce website is a typical example of what kind of generic application architecture? [1]

Transaction processing system

f. Which Gang of Four design pattern is best suited to uniformly handling the user manipulations of a user interface window containing a number of components? [1]

Composite pattern

g. Indicate two ways in which open source has impacted on society. [2]

Undermines the idea that work should be done for economic/financial gain

Challenges traditional notions of copyright and intellectual property, which restricts others from modifying "creative" work

Open Source software is used extensively in ICT for Development (ICT4D) since no or little cost, as compared to proprietary software

Creative Commons licensing scheme for creative works inspired by Open Source

Other areas of production also inspired by Open Source, e.g. Open Educational Resources, Open Source Hardware,

//any two [1 mark each]

Section 2. Theory of Algorithms [11 Marks]

Question 4 [11 Marks]

a. Brute force and divide-/decrease-and-conquer are techniques most commonly used for problem solving. Name and briefly describe ***any two other*** techniques. [2]

Space-time: use additional data structures/memory/variables/space in order to make a faster solution/algorithm possible.

Dynamic programming: solve overlapping subproblems by storing intermediate results in an array where they are later accessed instead of being recomputed

Transform and conquer: solve a problem more easily by changing it to a more convenient instance of the same problem, or using a different representation, or transforming it to another problem whose solution can be transformed into an answer to the original (any 1 of the three will do)

Greedy: solve a problem by making the optimal choice at each step

b. Consider the Master Theorem below and answer the questions that follow: [3]

If $T(n) = aT(n/b) + f(n)$ then $T(n)$ is $O(n^d)$ if $a < b^d$

$T(n)$ is $O(n^d \log n)$ if $a = b^d$

$T(n)$ is $O(n^{\log_b a})$ if $a > b^d$

i. Which letter indicates how much recursion an algorithm does: a, b or d?

ii. Which indicates how many operations each recursive call does: a, b or d?

iii. What does the remaining letter indicate?

a

d

what fraction of the input is the argument to each recursive call

c. Answer ***either one*** of the following questions: [4]

i. Show the ***shift table*** that would be used by the Boyer-Moore and Horspool algorithms when searching for the string DESEEDED in a text; and state by ***how many positions each algorithm would shift*** the pattern (DESEEDED) after their very first attempt i.e. when they fail to find that string at the start of the text "HE-FOUND-AND-DESEEDED-IT" (state clearly which answer is BM and which Horspool).

Array: D = 2; E = 1; S = 5. Horspool moves 2; BM moves 7.

ii. The initial array A(1) of Floyd's algorithm for finding shortest paths is shown below (∞ represents infinity). Show the ***next array A(2)*** that would result, and show what ***the other array*** used by this algorithm would contain at that stage i.e. when A(2) has been constructed, what does the other array contain?

	1	2	3	4
1	**0**	∞	**3**	∞
2	**2**	**0**	∞	∞
3	∞	**7**	**0**	**1**
4	**6**	∞	∞	**0**

Solution (where "any" can be 0 or null any value other than 1, 2, 3 or 4):

	1	2	3	4
1	**any**	**any**	**any**	**any**
2	**any**	**any**	**1**	**any**
3	**any**	**any**	**any**	**any**
4	**any**	**any**	**1**	**any**

	1	2	3	4
1	**0**	∞	**3**	∞
2	**2**	**0**	**5**	∞
3	∞	**7**	**0**	**1**
4	**6**	∞	**9**	**0**

d. You have N rolls of material, all 1 metre wide but all of different lengths. You need to cut K strips of material all 1 metre wide and want each of the K strips to be the same length. Given K and N, and the length of each of the N rolls, how do you find the maximum length your K strips can be? You can NOT join material from different rolls together, you can only cut. Assume K and N are at most 100.

Outline briefly in words (about 1 line each) two (2) algorithms that have different efficiency classes, and both solve this problem, stating which is more efficient. [2]

Try each possible length in turn going from 1 to 100 (or from 100 to 1)

Do a binary search, which is more efficient than the above.

Section 3. Theory of Computation [8 Marks]

Question 5 [8 Marks]

a. Let N be an NFA, and let N' be an NFA obtained from it by letting the accept states of N' be the reject states of N, and the reject states of N' the accept states of N (so N' is obtained from N by swapping around the accept and reject states). Is the language accepted by N' the complement of the language accepted by N? If it is, explain why it is. If it is not, give an example to show that it isn't. [3]

No it's not. Many examples. Consider the following NFA N over {0,1}. It accepts the language {0}. N' accepts the language {ε}, which is not the complement of L(N). The problem cases are those where the NFA gets stuck



b. Give a regular expression that accepts the following language: [2]

L = { w | w has an odd number of 1s, and ends with a 0}.

*One of many options: 0*10*(0*10*10*)*0*

c. Complete the following proof to prove that the following language is not regular: [3]

L = { $a^n b^n a^n$ | n≥0}.

Proof: Assume L is regular and consider the word w = …… where p is the pumping length. By the pumping lemma, w must be a word of the form xyz where the length of xy is no greater than p and y has a length greater than 0. This means that the word xy^2z is not a word in L because…….: a contradiction.

Assume L is regular and consider the word w = $a^p b^p a^p$ where p is the pumping length. By the pumping lemma, w must be a word of the form xyz where the length of xy is greater than p, and y has a length greater than 0. This means that the word xy^2z is not a word in L because it has the form $a^q b^p a^p$ where q>p (since xy consists of 0s only): a contradiction.